DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-200-01 — Internal inspection of vertical vessel C-101

Area	A200 Vessels and column	**Task**	VI Internal inspection
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	C-101 — C-101 vertical vessel 48" ID × 240" T/T — E 68'-6" N 25'-0" EL 100'-8-1/4"		
Equipment	C-101	**Lines**	4"-P300-P-1010, 6"-P300-P-1011, 6"-P300-P-1017, 3"-P300-P-1015, 1"-P150-V-6011, 1"-P150-D-6012, 1"-P150-N-3013, 3/4"-P300-I-6013, 3/4"-P300-I-6014, 3/4"-P300-SP-6015
Isometrics	ISO-P-1010, ISO-P-1011, ISO-P-1017, ISO-P-1015, ISO-V-6011, ISO-D-6012, ISO-N-3013, ISO-I-6013, ISO-I-6014, ISO-SP-6015 (owner's drawings, attached)		
Permits	General work permit, Confined-space entry permit		

Reason

Scheduled internal inspection of the column.

Scope of work

1. Isolate C-101 per the isolation list; drain, vent, purge and gas-test for entry.
2. Open for entry; clean as needed for inspection.
3. Internal visual examination; ultrasonic thickness at the 16 TMLs.
4. Close with new gaskets on every opened joint; remove blinds; restore the line-up.

Affected items: C-101

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-6, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1010	4"-P300-P-1010	4"-P300-P-1010	4	agrees with the model
ISO-P-1011	6"-P300-P-1011	6"-P300-P-1011	2	agrees with the model
ISO-P-1017	6"-P300-P-1017	6"-P300-P-1017	4	agrees with the model
ISO-P-1015	3"-P300-P-1015	3"-P300-P-1015	6	agrees with the model
ISO-V-6011	1"-P150-V-6011	1"-P150-V-6011	1	agrees with the model
ISO-D-6012	1"-P150-D-6012	1"-P150-D-6012	1	agrees with the model
ISO-N-3013	1"-P150-N-3013	1"-P150-N-3013	1	agrees with the model
ISO-I-6013	3/4"-P300-I-6013	3/4"-P300-I-6013	1	agrees with the model
ISO-I-6014	3/4"-P300-I-6014	3/4"-P300-I-6014	1	agrees with the model
ISO-SP-6015	3/4"-P300-SP-6015	3/4"-P300-SP-6015	1	agrees with the model

ISO-P-1010 (PDF): ../isometrics-imported/ISO-P-1010.pdf · ISO-P-1011 (PDF): ../isometrics-imported/ISO-P-1011.pdf · ISO-P-1017 (PDF): ../isometrics-imported/ISO-P-1017.pdf · ISO-P-1015 (PDF): ../isometrics-imported/ISO-P-1015.pdf · ISO-V-6011 (PDF): ../isometrics-imported/ISO-V-6011.pdf · ISO-D-6012 (PDF): ../isometrics-imported/ISO-D-6012.pdf · ISO-N-3013 (PDF): ../isometrics-imported/ISO-N-3013.pdf · ISO-I-6013 (PDF): ../isometrics-imported/ISO-I-6013.pdf · ISO-I-6014 (PDF): ../isometrics-imported/ISO-I-6014.pdf · ISO-SP-6015 (PDF): ../isometrics-imported/ISO-SP-6015.pdf — the sheets that hold the work follow the ITP.

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
C-101	equipment	C-101 vertical vessel 48" ID × 240" T/T		E 68'-6" N 25'-0" EL 100'-8-1/4"
4"-P300-P-1010/B2	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1010	E 68'-6" N 21'-8-3/16" EL 116'-0"
C-101-N1	nozzle	Nozzle C-101-N1 (feed, 4" P300) on C-101		E 68'-6" N 22'-3-15/16" EL 116'-0"
6"-P300-P-1011/A2	item	VALVE, GATE, WEDGE, OS&Y 6" P300	6"-P300-P-1011	E 68'-6" N 25'-0" EL 127'-11-11/16"
C-101-N2	nozzle	Nozzle C-101-N2 (overhead, 6" P300) on C-101		E 68'-6" N 25'-0" EL 127'-4-9/16"
6"-P300-P-1017/A2	item	VALVE, GATE, WEDGE, OS&Y 6" P300	6"-P300-P-1017	E 68'-6" N 21'-8-9/16" EL 106'-6"
C-101-N3	nozzle	Nozzle C-101-N3 (bottoms, 6" P300) on C-101		E 68'-6" N 22'-3-3/4" EL 106'-6"
3"-P300-P-1015/B2	item	VALVE, GATE, WEDGE, OS&Y 3" P300	3"-P300-P-1015	E 68'-6" N 21'-9-3/4" EL 125'-0"
C-101-N4	nozzle	Nozzle C-101-N4 (reflux, 3" P300) on C-101		E 68'-6" N 22'-4-11/16" EL 125'-0"
1"-P150-V-6011/2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-V-6011	E 70'-0" N 25'-0" EL 127'-4-13/16"
C-101-N5	nozzle	Nozzle C-101-N5 (vent, 1" P150) on C-101		E 70'-0" N 25'-0" EL 126'-9-15/16"
1"-P150-D-6012/2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-D-6012	E 65'-5-1/2" N 25'-0" EL 106'-0"
C-101-N6	nozzle	Nozzle C-101-N6 (drain, 1" P150) on C-101		E 66'-3/8" N 25'-0" EL 106'-0"

Tag	Kind	Description	Line	Location
1"-P150-N-3013/B2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-N-3013	E 68'-6" N 28'-1/2" EL 115'-0"
C-101-N7	nozzle	Nozzle C-101-N7 (nitrogen, 1" P150) on C-101		E 68'-6" N 27'-5-11/16" EL 115'-0"
3/4"-P300-I-6013/2	item	VALVE, BALL, FLOATING 3/4" P300	3/4"-P300-I-6013	E 65'-5-3/4" N 25'-0" EL 108'-6"
C-101-N8	nozzle	Nozzle C-101-N8 (instrument, 0.75" P300) on C-101		E 66'-1/2" N 25'-0" EL 108'-6"
3/4"-P300-I-6014/2	item	VALVE, BALL, FLOATING 3/4" P300	3/4"-P300-I-6014	E 68'-6" N 28'-1/4" EL 117'-6"
C-101-N9	nozzle	Nozzle C-101-N9 (instrument, 0.75" P300) on C-101		E 68'-6" N 27'-5-9/16" EL 117'-6"
3/4"-P300-SP-6015/2	item	VALVE, BALL, FLOATING 3/4" P300	3/4"-P300-SP-6015	E 65'-5-3/4" N 25'-0" EL 107'-6"
C-101-N10	nozzle	Nozzle C-101-N10 (sample, 0.75" P300) on C-101		E 66'-1/2" N 25'-0" EL 107'-6"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	4"-P300-P-1010/B2	4"-P300-P-1010	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1010/B2) on C-101-N1	E 68'-6" N 21'-8-3/16" EL 116'-0"		
2	blind	C-101-N1	4"-P300-P-1010	Fit a blind between 4"-P300-P-1010/B2 and the nozzle flange C-101-N1	E 68'-6" N 22'-3-15/16" EL 116'-0"		
3	valve	6"-P300-P-1011/A2	6"-P300-P-1011	Close and lock VALVE, GATE, WEDGE, OS&Y 6" P300 (6"-P300-P-1011/A2) on C-101-N2	E 68'-6" N 25'-0" EL 127'-11-11/16"		
4	blind	C-101-N2	6"-P300-P-1011	Fit a blind between 6"-P300-P-1011/A2 and the nozzle flange C-101-N2	E 68'-6" N 25'-0" EL 127'-4-9/16"		
5	valve	6"-P300-P-1017/A2	6"-P300-P-1017	Close and lock VALVE, GATE, WEDGE, OS&Y 6" P300 (6"-P300-P-1017/A2) on C-101-N3	E 68'-6" N 21'-8-9/16" EL 106'-6"		
6	blind	C-101-N3	6"-P300-P-1017	Fit a blind between 6"-P300-P-1017/A2 and the nozzle flange C-101-N3	E 68'-6" N 22'-3-3/4" EL 106'-6"		
7	valve	3"-P300-P-1015/B2	3"-P300-P-1015	Close and lock VALVE, GATE, WEDGE, OS&Y 3" P300 (3"-P300-P-1015/B2) on C-101-N4	E 68'-6" N 21'-9-3/4" EL 125'-0"		
8	blind	C-101-N4	3"-P300-P-1015	Fit a blind between 3"-P300-P-1015/B2 and the nozzle flange C-101-N4	E 68'-6" N 22'-4-11/16" EL 125'-0"		
9	valve	1"-P150-V-6011/2	1"-P150-V-6011	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-V-6011/2) on C-101-N5	E 70'-0" N 25'-0" EL 127'-4-13/16"		
10	blind	C-101-N5	1"-P150-V-6011	Fit a blind between 1"-P150-V-6011/2 and the nozzle flange C-101-N5	E 70'-0" N 25'-0" EL 126'-9-15/16"		
11	valve	1"-P150-D-6012/2	1"-P150-D-6012	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-D-6012/2) on C-101-N6	E 65'-5-1/2" N 25'-0" EL 106'-0"		
12	blind	C-101-N6	1"-P150-D-6012	Fit a blind between 1"-P150-D-6012/2 and the nozzle flange C-101-N6	E 66'-3/8" N 25'-0" EL 106'-0"		
13	valve	1"-P150-N-3013/B2	1"-P150-N-3013	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-N-3013/B2) on C-101-N7	E 68'-6" N 28'-1/2" EL 115'-0"		
14	blind	C-101-N7	1"-P150-N-3013	Fit a blind between 1"-P150-N-3013/B2 and the nozzle flange C-101-N7	E 68'-6" N 27'-5-11/16" EL 115'-0"		
15	valve	3/4"-P300-I-6013/2	3/4"-P300-I-6013	Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-I-6013/2) on C-101-N8	E 65'-5-3/4" N 25'-0" EL 108'-6"		
16	blind	C-101-N8	3/4"-P300-I-6013	Fit a blind between 3/4"-P300-I-6013/2 and the nozzle flange C-101-N8	E 66'-1/2" N 25'-0" EL 108'-6"		
17	valve	3/4"-P300-I-6014/2	3/4"-P300-I-6014	Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-I-6014/2) on C-101-N9	E 68'-6" N 28'-1/4" EL 117'-6"		
18	blind	C-101-N9	3/4"-P300-I-6014	Fit a blind between 3/4"-P300-I-6014/2 and the nozzle flange C-101-N9	E 68'-6" N 27'-5-9/16" EL 117'-6"		
19	valve	3/4"-P300-SP-6015/2	3/4"-P300-SP-6015	Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-SP-6015/2) on C-101-N10	E 65'-5-3/4" N 25'-0" EL 107'-6"		
20	blind	C-101-N10	3/4"-P300-SP-6015	Fit a blind between 3/4"-P300-SP-6015/2 and the nozzle flange C-101-N10	E 66'-1/2" N 25'-0" EL 107'-6"		

Inspections

Id	Inspection	Criteria	Basis
WO-200-01-I1	Internal visual examination of the shell, heads and nozzles	Findings recorded against the owner's integrity criteria	SW-NDE-4
WO-200-01-I2	Ultrasonic thickness at the 16 TMLs listed	Each reading above the owner's minimum wall	SW-WO-6

Thickness monitoring locations (SW-WO-6)

TML	Item	Location	Position	Nominal wall	R1	R2	R3	R4
WO-200-01-TML01	C-101	Nozzle neck C-101-N1	E 68'-6" N 22'-3-15/16" EL 116'-0"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML02	C-101	Nozzle neck C-101-N2	E 68'-6" N 25'-0" EL 127'-4-9/16"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML03	C-101	Nozzle neck C-101-N3	E 68'-6" N 22'-3-3/4" EL 106'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML04	C-101	Nozzle neck C-101-N4	E 68'-6" N 22'-4-11/16" EL 125'-0"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML05	C-101	Nozzle neck C-101-N5	E 70'-0" N 25'-0" EL 126'-9-15/16"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML06	C-101	Nozzle neck C-101-N6	E 66'-3/8" N 25'-0" EL 106'-0"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML07	C-101	Nozzle neck C-101-N7	E 68'-6" N 27'-5-11/16" EL 115'-0"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML08	C-101	Nozzle neck C-101-N8	E 66'-1/2" N 25'-0" EL 108'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML09	C-101	Nozzle neck C-101-N9	E 68'-6" N 27'-5-9/16" EL 117'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML10	C-101	Nozzle neck C-101-N10	E 66'-1/2" N 25'-0" EL 107'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-200-01-TML11	C-101	Bottom head centre	E 68'-6" N 25'-0" EL 103'-8"		centre: ____			
WO-200-01-TML12	C-101	Top head centre	E 68'-6" N 25'-0" EL 125'-8-1/2"		centre: ____			
WO-200-01-TML13	C-101	Shell at 0.000" above the lower tangent	E 68'-6" N 25'-0" EL 104'-8-1/4"		0°: ____	90°: ____	180°: ____	270°: ____
WO-200-01-TML14	C-101	Shell at 6'-0" above the lower tangent	E 68'-6" N 25'-0" EL 110'-8-1/4"		0°: ____	90°: ____	180°: ____	270°: ____
WO-200-01-TML15	C-101	Shell at 12'-0" above the lower tangent	E 68'-6" N 25'-0" EL 116'-8-1/4"		0°: ____	90°: ____	180°: ____	270°: ____
WO-200-01-TML16	C-101	Shell at 18'-0" above the lower tangent	E 68'-6" N 25'-0" EL 122'-8-1/4"		0°: ____	90°: ____	180°: ____	270°: ____

Test — Service leak test

Description	Every opened joint checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

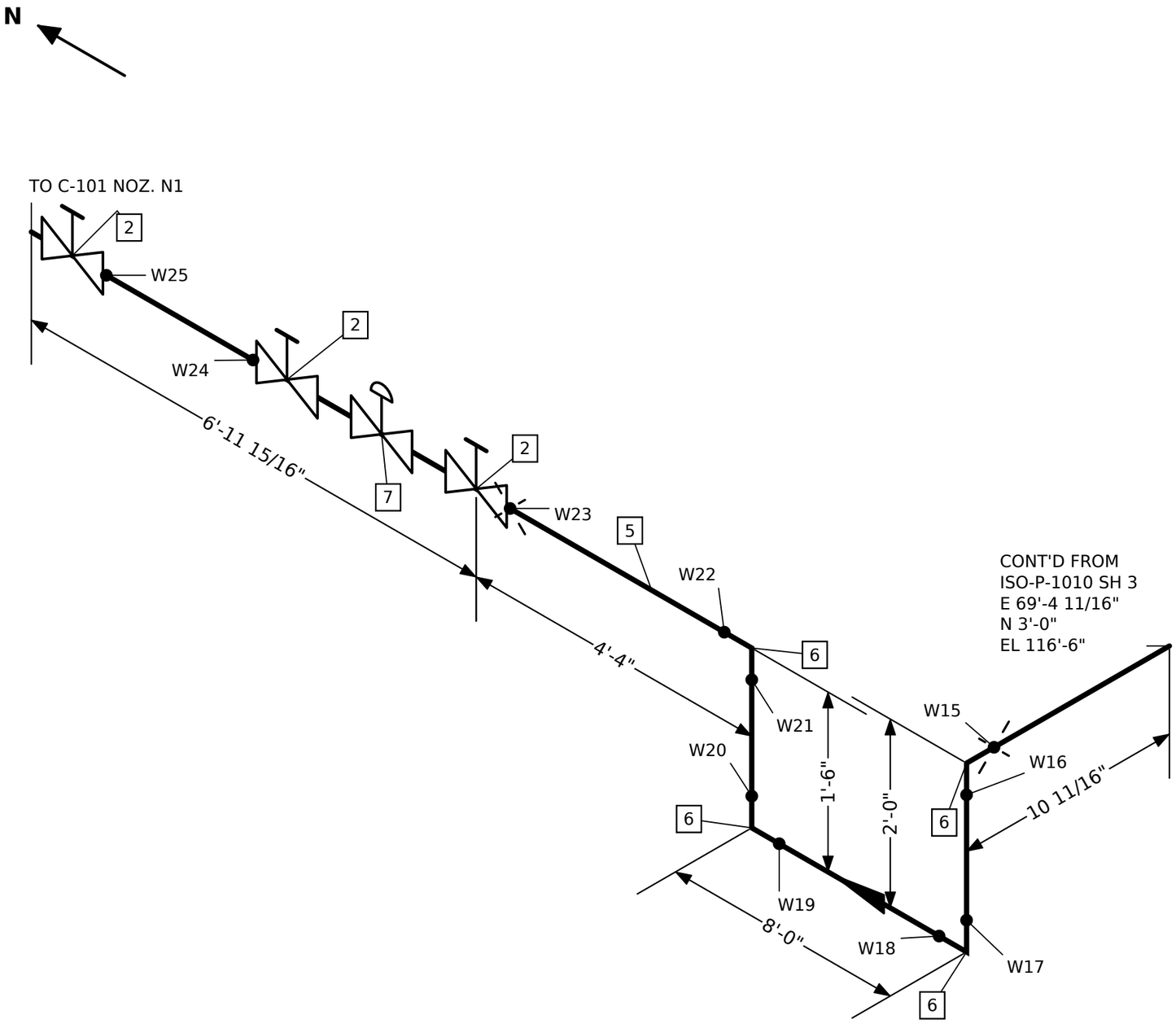
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Equipment vented, purged and gas-tested for entry; confined-space entry permit issued	Owner entry procedure	Atmosphere safe for entry per owner limits	Gas test record, entry permit	H	H	—				
3	Opened for entry; cleaned for inspection	Owner procedure	Surfaces clean enough to inspect	Daily report	S	S	—				
4	Internal visual examination	SW-NDE-4	Findings recorded	Inspection report with photographs	W	H	W				
5	Ultrasonic thickness at 16 TMLs	SW-WO-6	Above the owner's minimum wall	Thickness record per TML	S	R	R				
6	Owner's decision on findings (repair, monitor or accept)	Owner integrity procedure	Decision recorded	Signed disposition	R	H	R				
7	Closed with new gaskets on every opened joint; studs to preload	SW-WO-5	Preload as stated	Bolt-up record	W	W	—				
8	Service leak test of opened joints on start-up	SW-TEST-2	No leak	Leak test record	W	W	—				
9	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
10	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

Tag	Role	Drawing	Sheet	Weld label
4"-P300-P-1010/B2	isolation: valve	ISO-P-1010	4	
6"-P300-P-1011/A2	isolation: valve	ISO-P-1011	1	
6"-P300-P-1017/A2	isolation: valve	ISO-P-1017	1	
3"-P300-P-1015/B2	isolation: valve	ISO-P-1015	6	
1"-P150-V-6011/2	isolation: valve	ISO-V-6011	1	
1"-P150-D-6012/2	isolation: valve	ISO-D-6012	1	
1"-P150-N-3013/B2	isolation: valve	ISO-N-3013	1	
3/4"-P300-I-6013/2	isolation: valve	ISO-I-6013	1	
3/4"-P300-I-6014/2	isolation: valve	ISO-I-6014	1	
3/4"-P300-SP-6015/2	isolation: valve	ISO-SP-6015	1	



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	6	4"	GASKET 4" P300 COMPRESSED SHEET
2	3	4"	VALVE, GATE, WEDGE, OS&Y 4" P300
3	6	4"	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300
4	3	4"	FLANGE WELD-NECK 4" P300 RAISED FACE
5	12'-8 3/16"	4"	PIPE 4" P300 BEVELLED ENDS
6	4	4"	ELBOW 90° R = 5/3 OD 4" P300
7	1	4"	VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

NOTES:

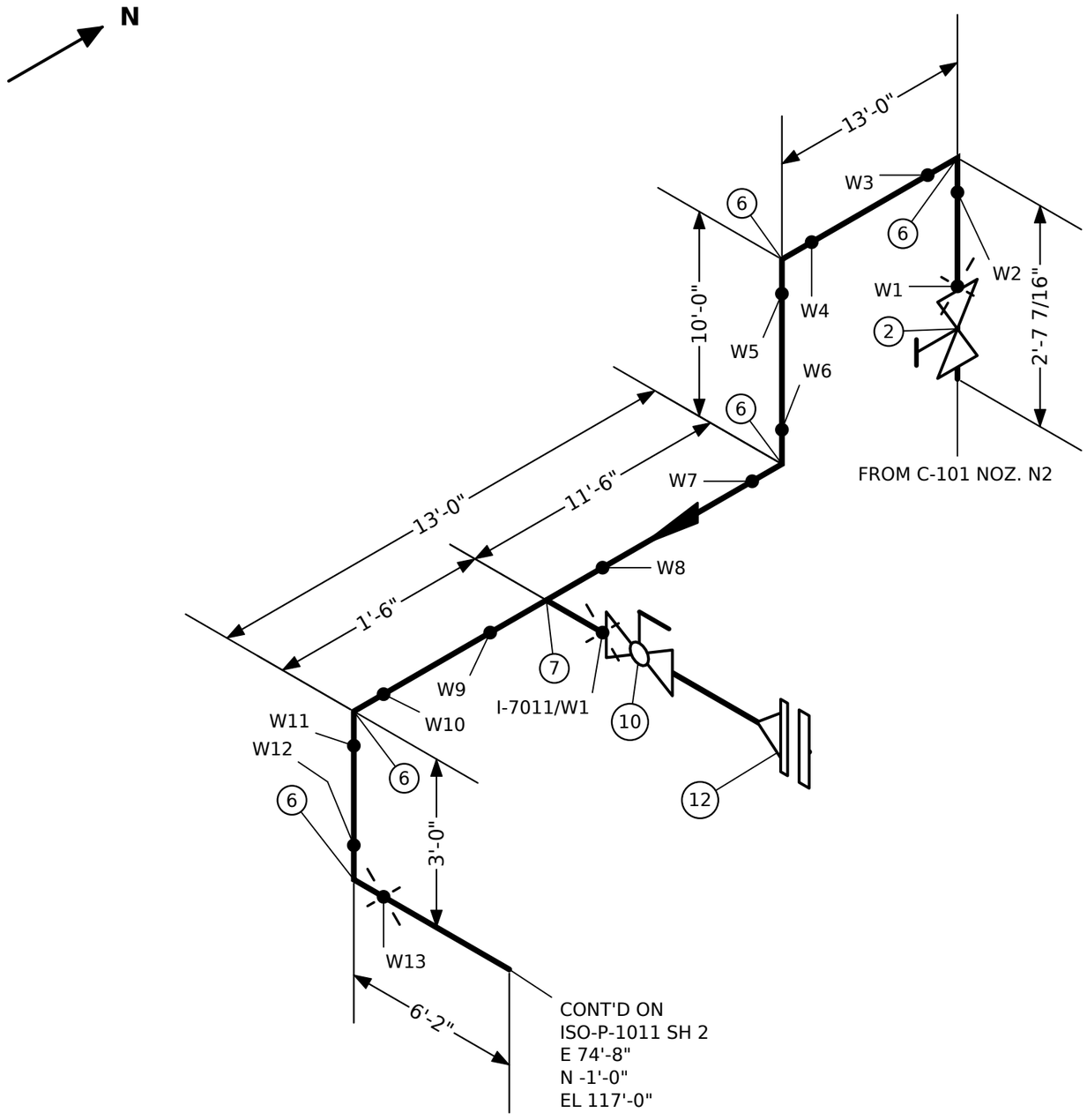
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	**NDE**	**FLUID**
-	-	P

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
4"-P300-P-1010	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1010	4 OF 4	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	6"	GASKET 6" P300 COMPRESSED SHEET
2	1	6"	VALVE, GATE, WEDGE, OS&Y 6" P300
3	2	6"	STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P300
4	1	6"	FLANGE WELD-NECK 6" P300 RAISED FACE
5	37'-1 1/4"	6"	PIPE 6" P300 BEVELLED ENDS
6	5	6"	ELBOW 90° R = 5/3 OD 6" P300
7	1	6"x3/4"	TEE REDUCING 6" × 3/4" P300
8	1	3/4"	FLANGE WELD-NECK 3/4" P300 RAISED FACE
9	2	3/4"	GASKET 3/4" P300 COMPRESSED SHEET
10	1	3/4"	VALVE, BALL, FLOATING 3/4" P300
11	2	3/4"	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300
12	1	3/4"	FLANGE BLIND 3/4" P300 RAISED FACE

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

NOTES:
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	**NDE**	**FLUID**
-	-	P

COMPANY

SMARTWELD

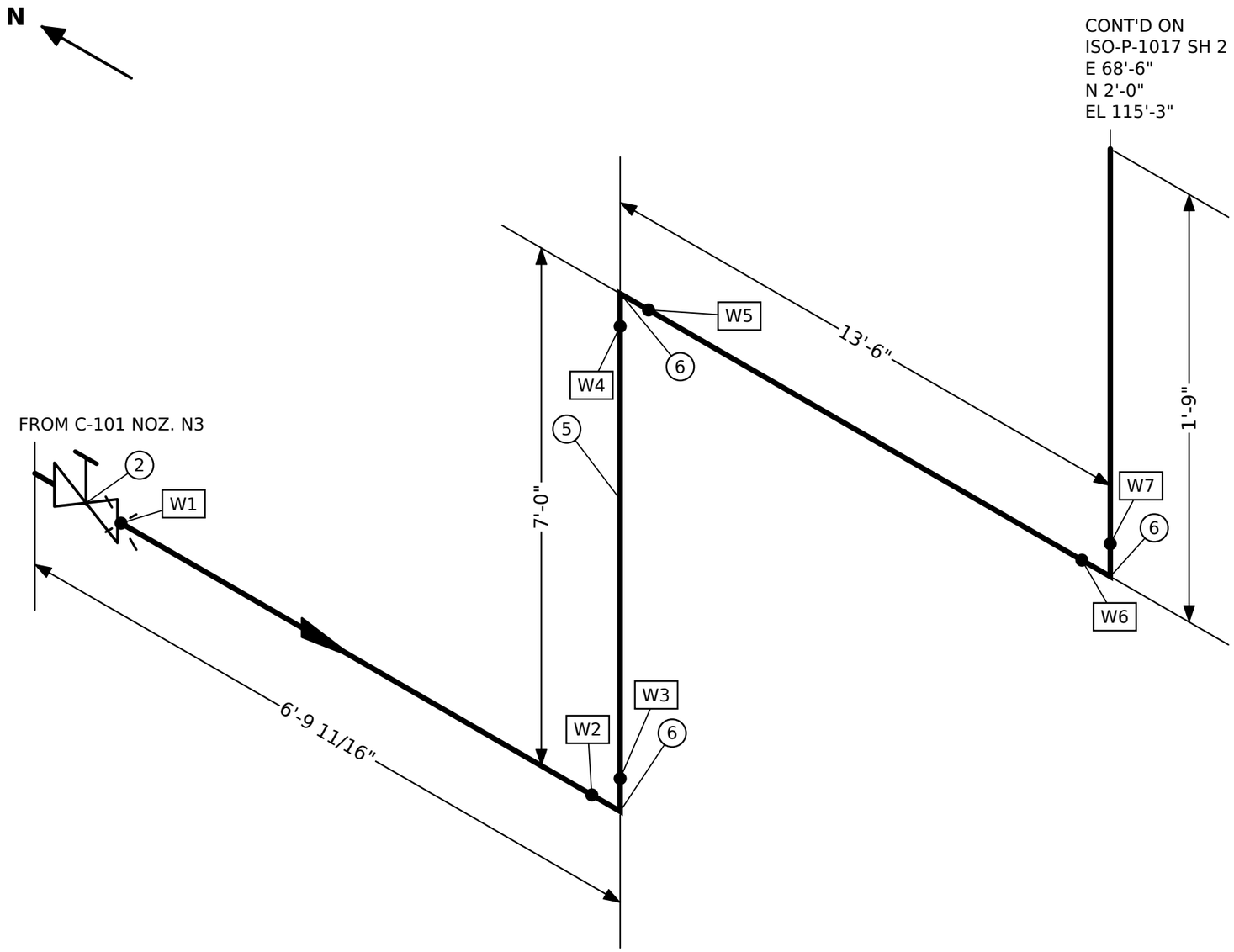
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HYDROCARBON PROCESS

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
6"-P300-P-1011	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1011	1 OF 2	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	6"	GASKET 6" P300 COMPRESSED SHEET
2	1	6"	VALVE, GATE, WEDGE, OS&Y 6" P300
3	2	6"	STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P300
4	1	6"	FLANGE WELD-NECK 6" P300 RAISED FACE
5	22'-6 1/2"	6"	PIPE 6" P300 BEVELLED ENDS
6	3	6"	ELBOW 90° R = 5/3 OD 6" P300

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. REMOVE BURRS FROM ALL CUT ENDS
4. HYDROTEST BEFORE PAINTING

TEST PRESS.	INSULATION	PAINT
-	NONE	-

PWHT	NDE	FLUID
-	-	P

COMPANY

SMARTWELD

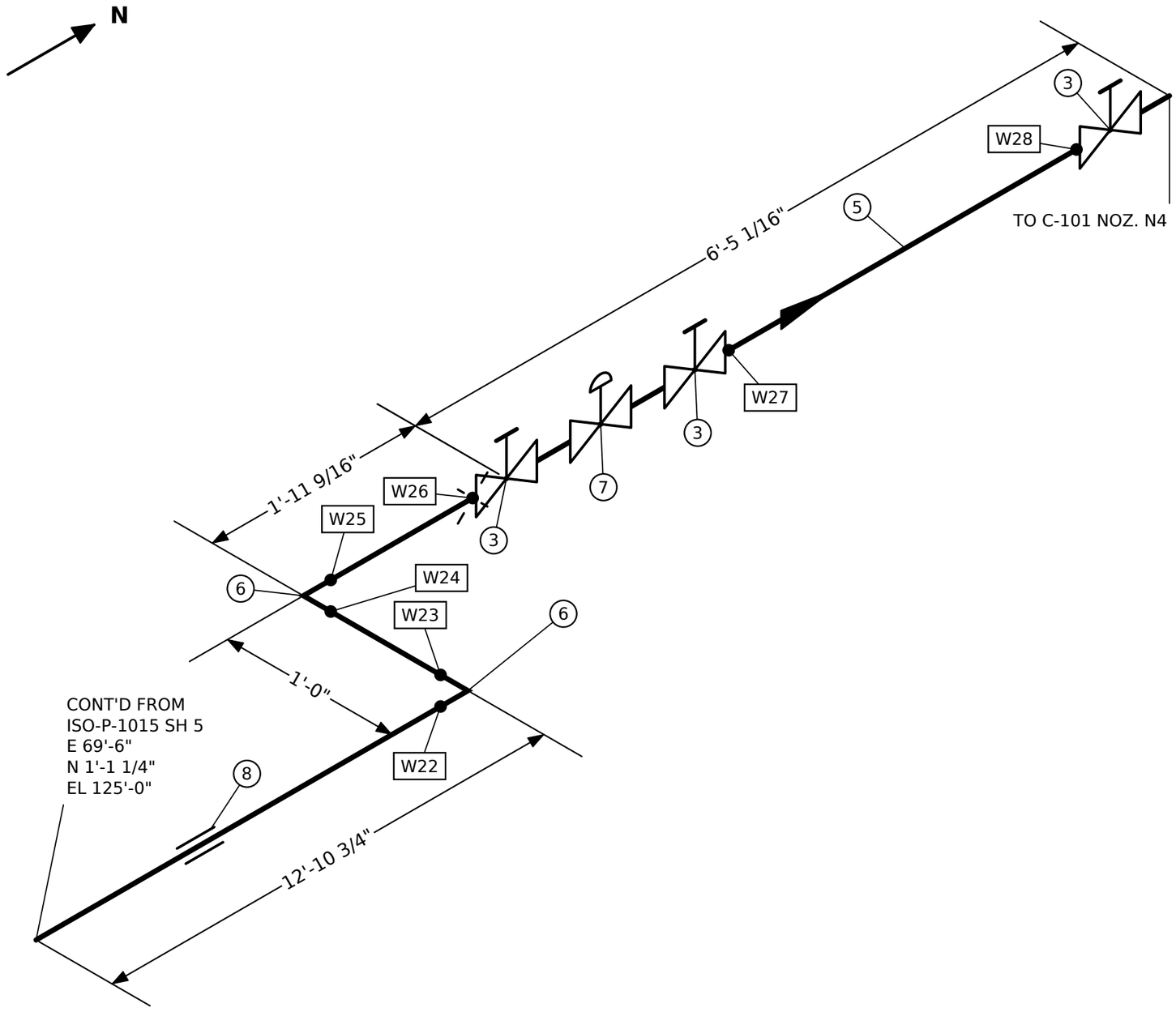
TITLE

HYDROCARBON PROCESS

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
6"-P300-P-1017	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1017	1 OF 4	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	6	3"	GASKET 3" P300 COMPRESSED SHEET
2	6	3"	STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300
3	3	3"	VALVE, GATE, WEDGE, OS&Y 3" P300
4	3	3"	FLANGE WELD-NECK 3" P300 RAISED FACE
5	14'-8 5/16"	3"	PIPE 3" P300 BEVELLED ENDS
6	2	3"	ELBOW 90° R = 5/3 OD 3" P300
7	1	3"	VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 3" P300
8	1	3"	HANGER FOR 3" LINE (HANGER-3x93.0)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

NOTES:

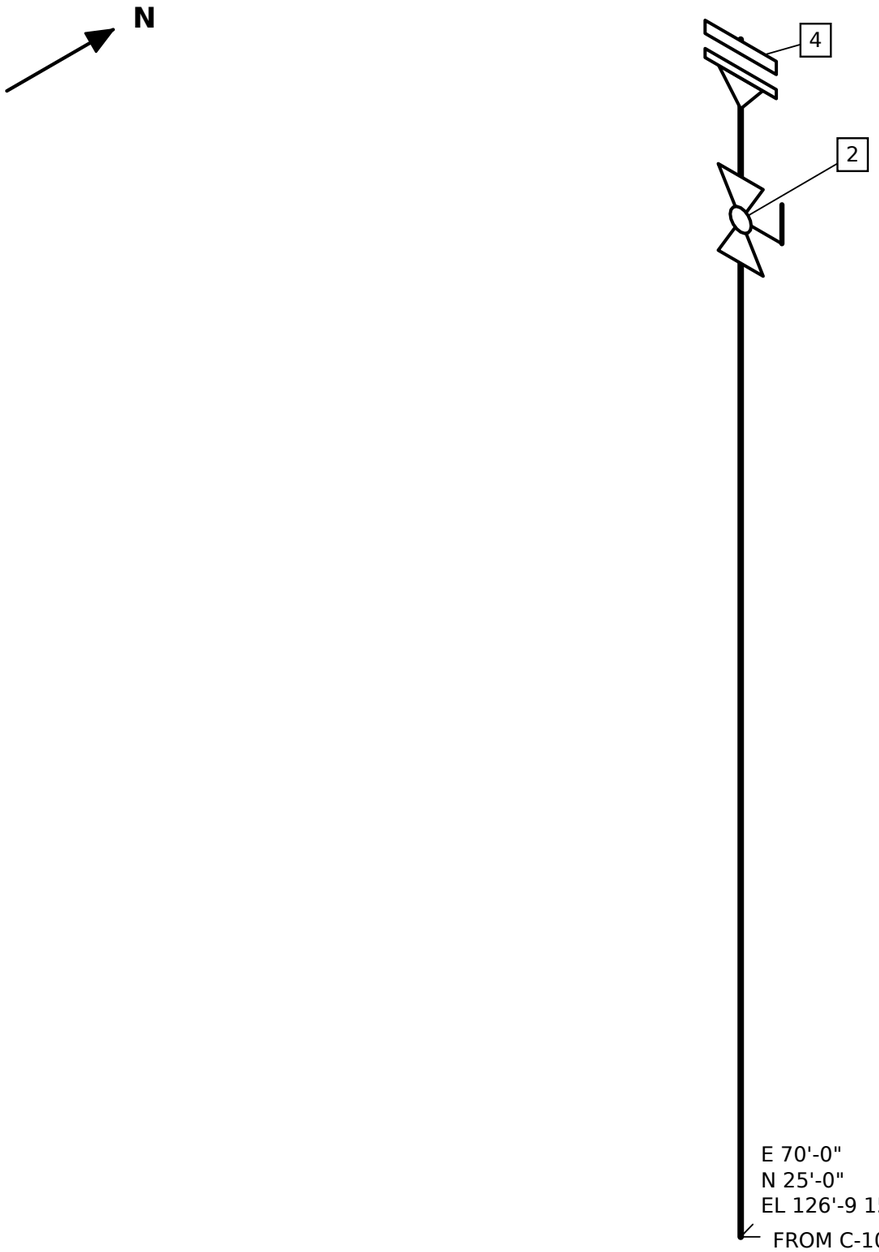
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. FLANGE BOLT HOLES STRADDLE CENTRELINES
4. PAINT PER PROJECT SPECIFICATION

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	P

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
3"-P300-P-1015	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1015	6 OF 6	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	1"	GASKET 1" P150 COMPRESSED SHEET
2	1	1"	VALVE, BALL, FLOATING 1" P150
3	2	1"	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150
4	1	1"	FLANGE BLIND 1" P150 RAISED FACE

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. PAINT PER PROJECT SPECIFICATION

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	**NDE**	**FLUID**
-	-	V

COMPANY

SMARTWELD

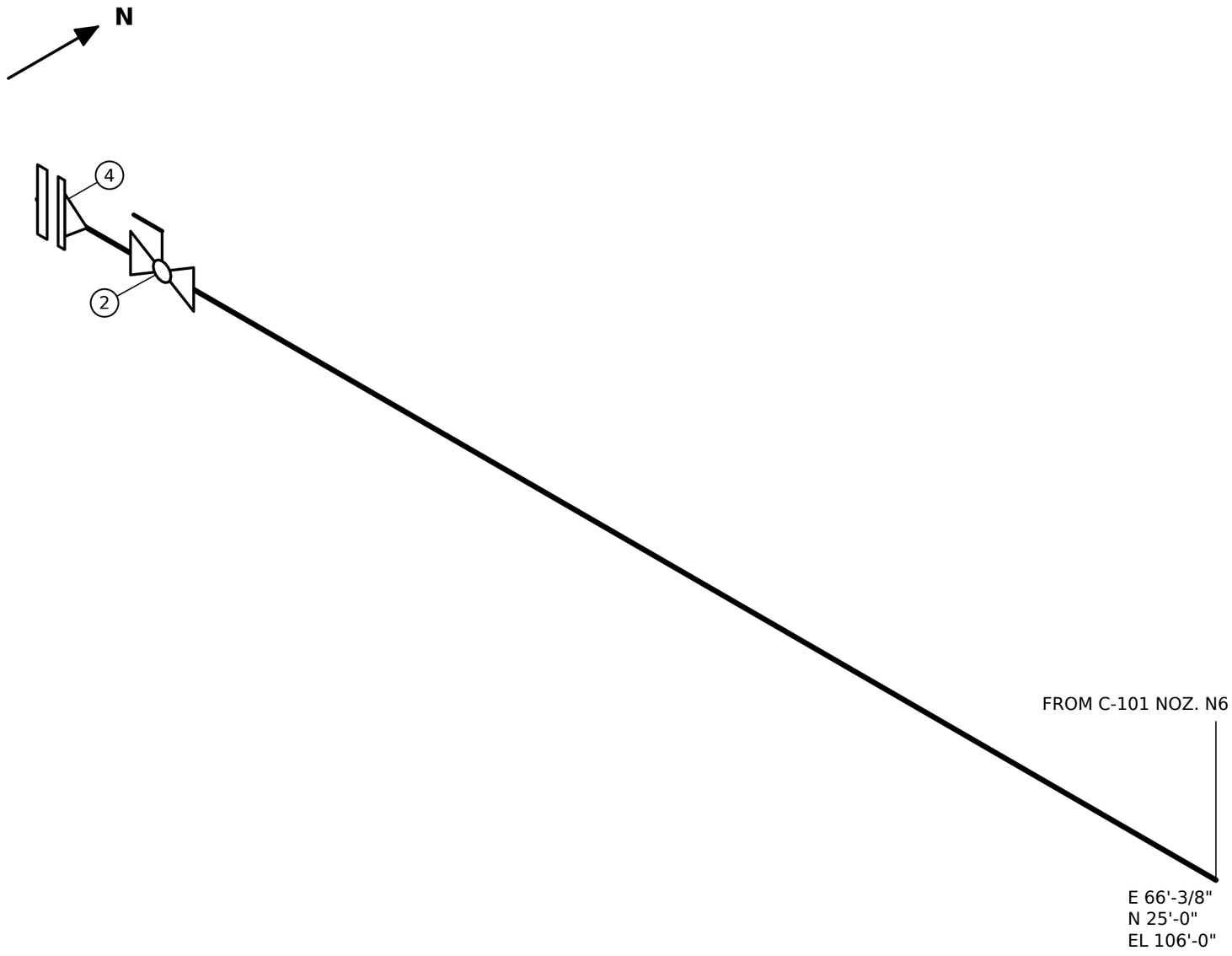
TITLE

HYDROCARBON PROCESS

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
1"-P150-V-6011	H150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-V-6011	1 OF 1	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	1"	GASKET 1" P150 COMPRESSED SHEET
2	1	1"	VALVE, BALL, FLOATING 1" P150
3	2	1"	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150
4	1	1"	FLANGE BLIND 1" P150 RAISED FACE

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. FLANGE BOLT HOLES STRADDLE CENTRELINES
4. PAINT PER PROJECT SPECIFICATION

TEST PRESS. -	INSULATION **NONE**	PAINT -	
PWHT -	NDE -	FLUID **D**	

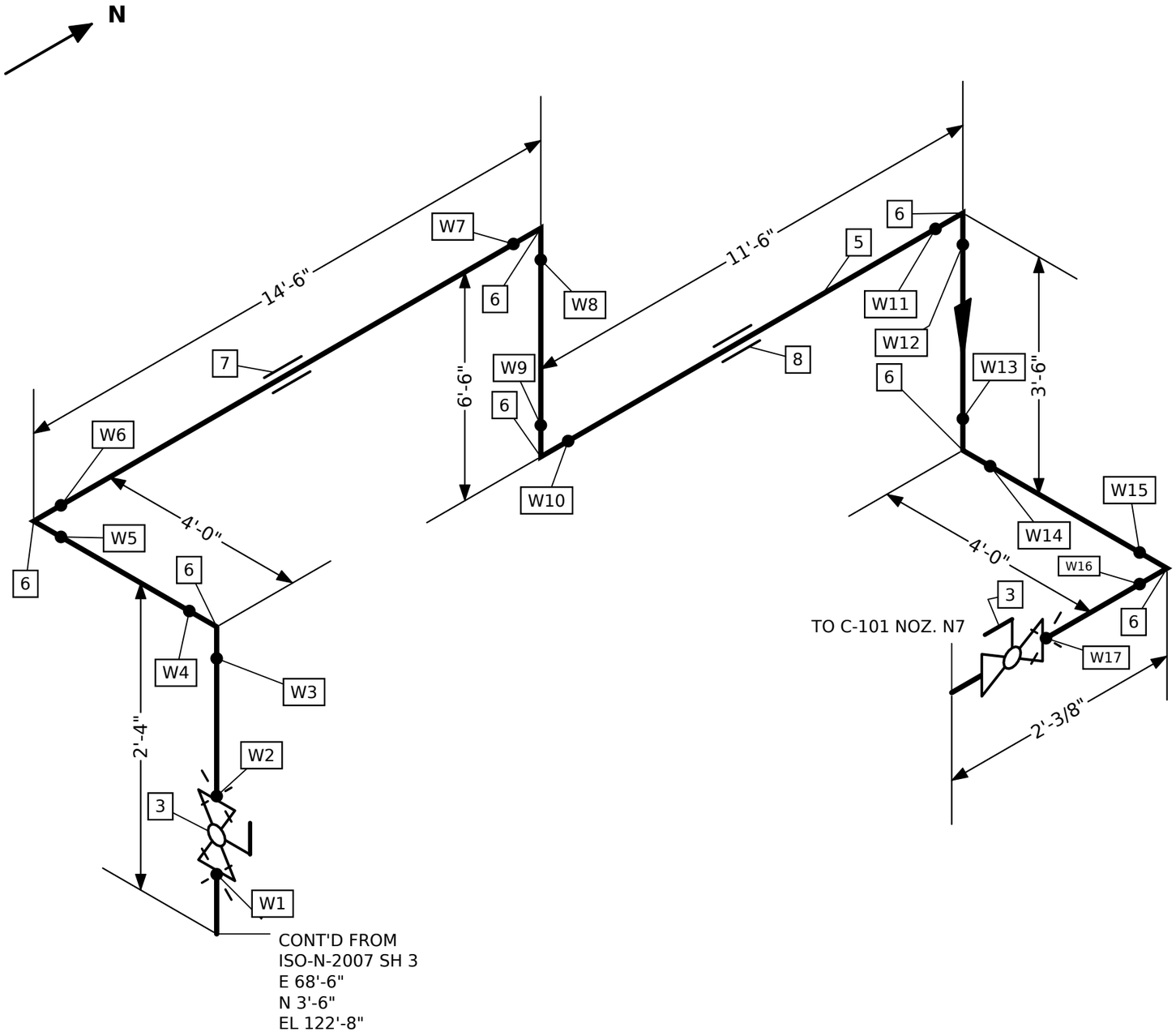
COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

LINE NO. **1"-P150-D-6012**	PIPING CLASS **H150**	DESIGN PRESS. **150 PSIG**	DESIGN TEMP. **-**
DRAWING NO. **ISO-D-6012**		SHEET **1 OF 1**	REV **1**
DATE **2026-11-01**	DRAWN **-**	CHECKED **-**	SCALE **NTS**



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	3	1"	FLANGE WELD-NECK 1" P150 RAISED FACE
2	4	1"	GASKET 1" P150 COMPRESSED SHEET
3	2	1"	VALVE, BALL, FLOATING 1" P150
4	4	1"	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150
5	43'-1 3/16"	1"	PIPE 1" P150 BEVELLED ENDS
6	7	1"	ELBOW 90° R = 5/3 OD 1" P150
7	1	1"	HANGER FOR 1" LINE (HANGER-1x88.0)
8	1	1"	HANGER FOR 1" LINE (HANGER-1x10.9)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

COMPANY

SMARTWELD

TITLE

AIR AND GASES

NOTES:

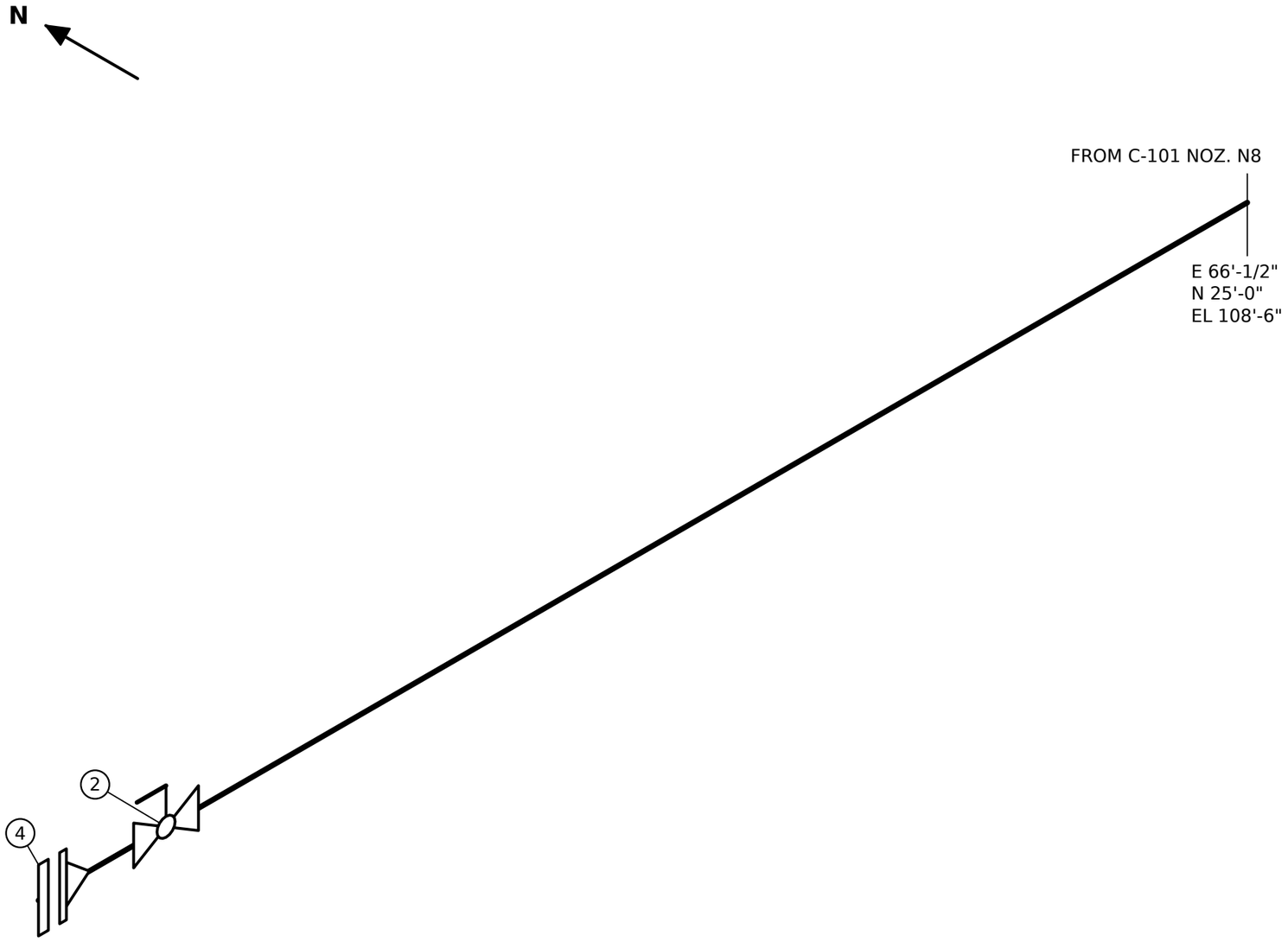
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	N

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
1"-P150-N-3013	G150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-N-3013	1 OF 1	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	3/4"	GASKET 3/4" P300 COMPRESSED SHEET
2	1	3/4"	VALVE, BALL, FLOATING 3/4" P300
3	2	3/4"	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300
4	1	3/4"	FLANGE BLIND 3/4" P300 RAISED FACE

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. CHECK ORIENTATION BEFORE WELDING

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	I

COMPANY

SMARTWELD

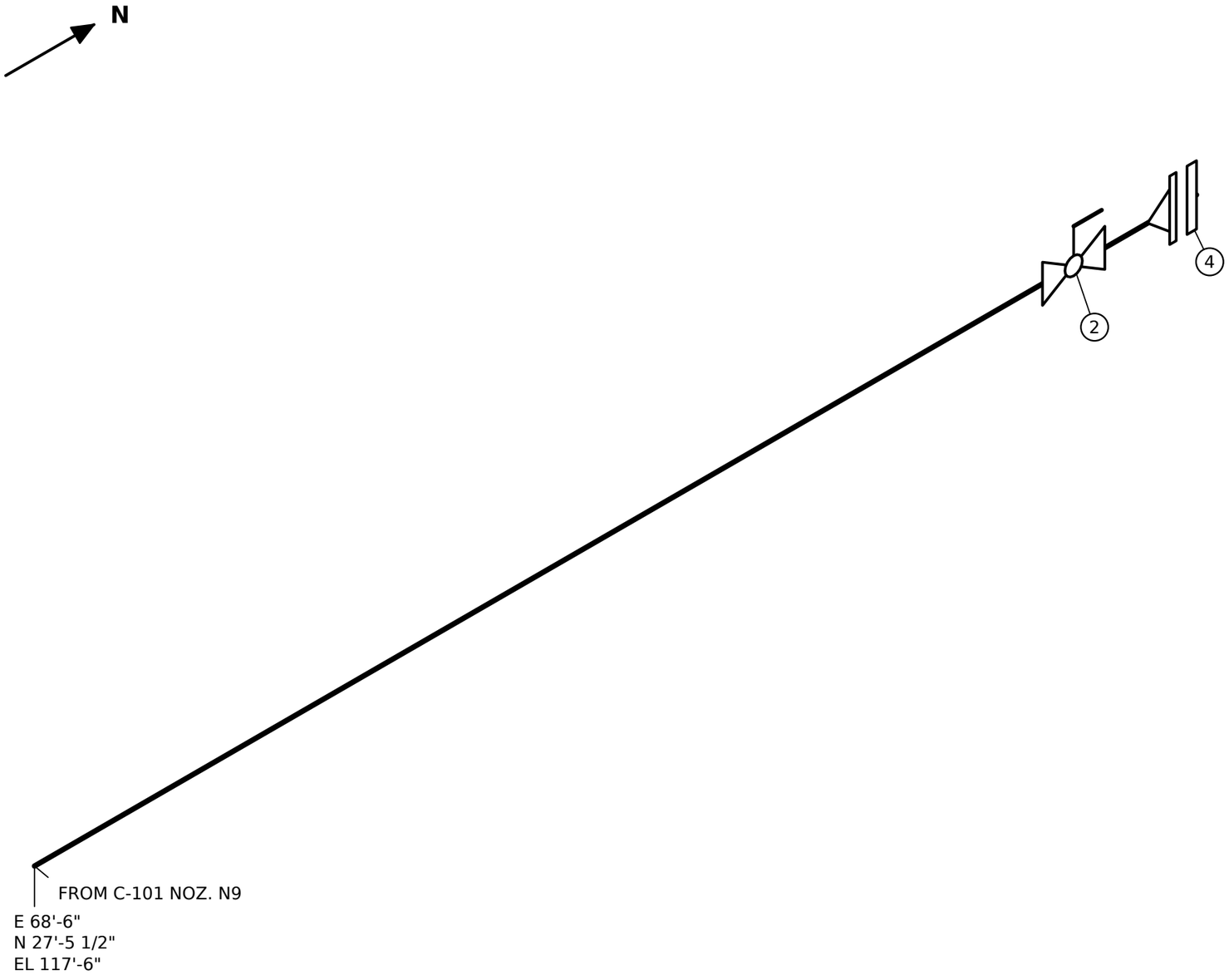
TITLE

HYDROCARBON PROCESS

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
3/4"-P300-I-6013	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-I-6013	1 OF 1	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	3/4"	GASKET 3/4" P300 COMPRESSED SHEET
2	1	3/4"	VALVE, BALL, FLOATING 3/4" P300
3	2	3/4"	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300
4	1	3/4"	FLANGE BLIND 3/4" P300 RAISED FACE

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. REMOVE BURRS FROM ALL CUT ENDS
4. HYDROTEST BEFORE PAINTING

TEST PRESS. -	INSULATION NONE	PAINT -
PWHT -	NDE -	FLUID I

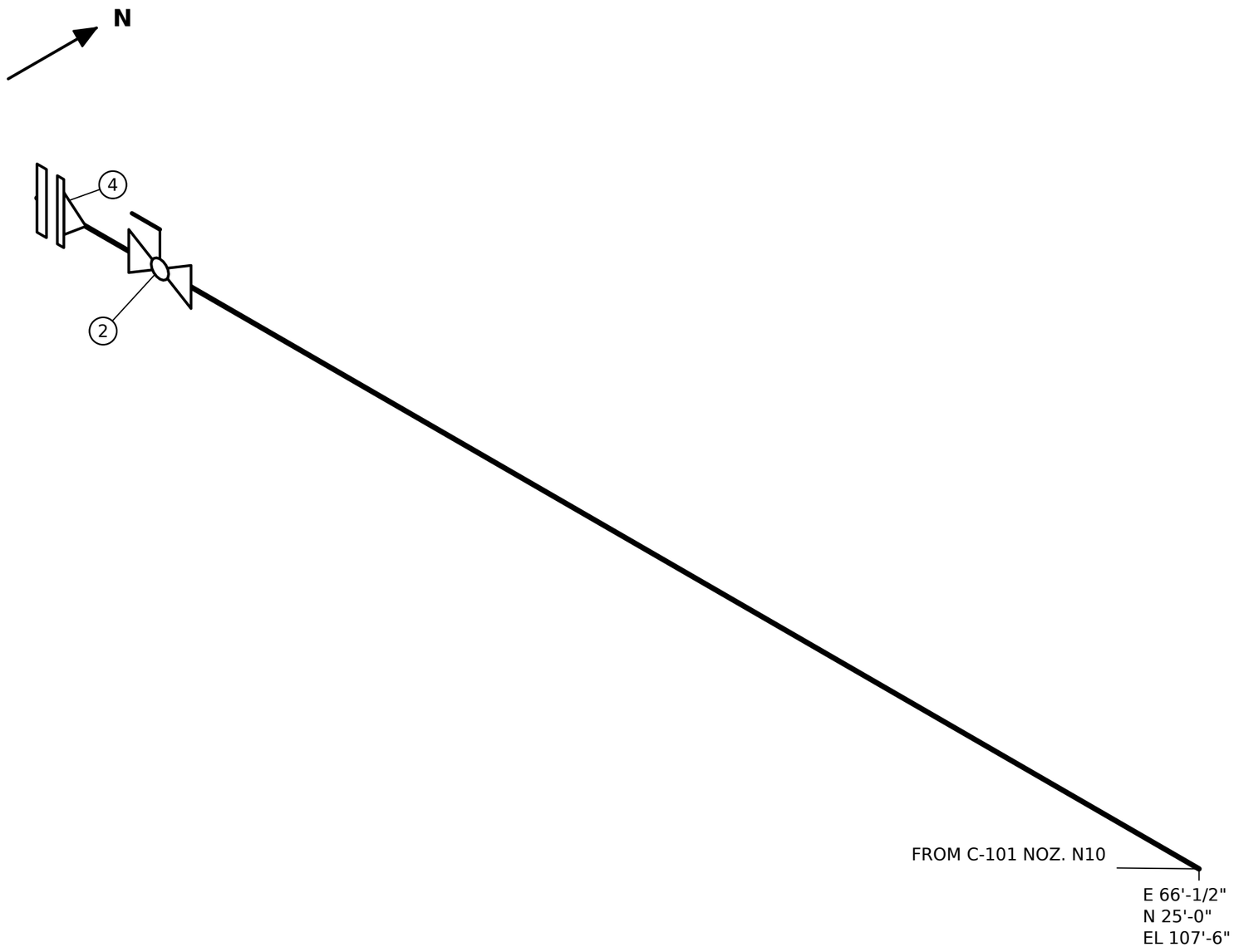
COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

LINE NO. 3/4"-P300-I-6014	PIPING CLASS H300	DESIGN PRESS. 300 PSIG	DESIGN TEMP. -
DRAWING NO. ISO-I-6014		SHEET 1 OF 1	REV 1
DATE 2026-11-01	DRAWN -	CHECKED -	SCALE NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	3/4"	GASKET 3/4" P300 COMPRESSED SHEET
2	1	3/4"	VALVE, BALL, FLOATING 3/4" P300
3	2	3/4"	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300
4	1	3/4"	FLANGE BLIND 3/4" P300 RAISED FACE

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. HYDROTEST BEFORE PAINTING

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	SP

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
3/4"-P300-SP-6015	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-SP-6015	1 OF 1	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS

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